

Chapter 7: A Tale of Three Intersecting Lines

Worksheet 7: A Tale of Three Intersecting Lines

Learning Objective: Classify triangles by sides and angles, and apply the angle sum property, the exterior angle theorem, and the triangle inequality.

Worked Example:

Two angles of a triangle are 55° and 65° . Find the third angle.

Solution: The angle sum of a triangle is 180° . Third angle = $180 - 55 - 65 = 60^\circ$.

Questions (1–20)

1. What is the sum of the interior angles of any triangle?
2. Classify a triangle with sides 5 cm, 5 cm, 5 cm.
3. Classify a triangle with angles 90° , 45° , 45° .
4. Can a triangle have two right angles? Why or why not?
5. Name the triangle with all sides of different lengths.
6. Two angles of a triangle are 40° and 80° . Find the third angle.

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7. An exterior angle of a triangle is 110° , and one interior opposite angle is 50° . Find the other interior opposite angle.

 8. Can a triangle have sides 3 cm, 4 cm and 8 cm? Use the triangle inequality to explain.

 9. A triangle has angles in the ratio 2:3:4. Find each angle.

 10. A triangle has angles 90° , x° , and 35° . Find x .

 11. An isosceles triangle has a vertex angle of 40° . Find the two equal base angles.

 12. The exterior angle of a triangle at one vertex is 125° . Find the sum of the two interior opposite angles, and if they are equal, find each.

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13. Check if a triangle can be formed with sides 7 cm, 10 cm and 2 cm. Explain using the triangle inequality.
14. In a triangle, one angle is twice another, and the third angle is 30° more than the smallest. Find all three angles.
15. A triangular park has angles 60° , 60° and 60° . What type of triangle is this, and what does it tell you about its sides?
16. The angles of a triangle are in the ratio 3:5:7. Is this triangle acute, right or obtuse? Find the angles.
17. Two sides of a triangle are 8 cm and 15 cm. Between what two values must the third side lie?
18. A triangle's exterior angle at vertex A is 100° , and at vertex B is 130° . Find all three interior angles.

19. Look at a roof, a cycle frame, or a hanger at home that uses a triangular shape. Why do you think triangles are used for strength in these structures?

20. Measure (or estimate) the three angles of a triangular object or drawing at home, and check if they add up to approximately 180° .

Reflection Question: Why do you think triangles are considered the most stable geometric shape, used in bridges and towers? Write your own idea.

Real-Life Application Question: Find a triangular object or design at home or school (a flag, a road sign, a roof truss). Try to classify it by its sides or angles.